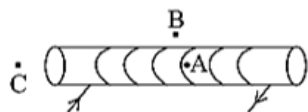


MAGNETISM -06

Class 10 - Science

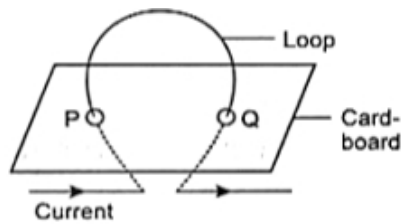
1. What is the function of an earth wire? Why is it necessary to earth metallic appliances? [2]
2. For the current carrying solenoid as shown, draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum? [2]



3. It is established that an electric current through a metallic conductor produces a magnetic field around it. Is there a similar magnetic field produced around a thin beam of moving
 - i. α -particles,
 - ii. neutrons? Justify your answer. [2]
4. Explain what is short circuiting and overloading in an electric supply ? [2]
5. How will you decide whether the magnetic field at a point is due to some current carrying conductor or due to earth? [2]
6. What is a solenoid? When does a solenoid behave as a magnet? Draw the pattern of the magnetic field produced inside it showing the directions of the magnetic field lines. [2]
7. Give a note on Magnetism in Human beings. [2]
8. Why does a current carrying freely suspended solenoid rest along a particular direction? State the direction in which it rests. [2]
9. Explain the meaning of overloading of an electrical circuit. List two possible causes due to which overloading may occur in household circuits. Write one preventive measure that should be taken to avoid overloading of domestic circuits. [2]
10. An electric switch should not be touched with wet hands while putting it on or off. Give a reason for your answer. [2]
11. What precaution should be taken to avoid the overloading of domestic electric circuits? [2]
12.
 - a. Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it. [2]
 - b. An alpha particle while passing through a magnetic field gets projected towards north. In which direction will an electron project when it passes through the same magnetic field?
13. Why does a compass needle get deflected when brought near a bar magnet? [2]
14. List two factors on which the strength of magnetic field at a point due to a current carrying straight conductor depends. State the rule that determines the direction of magnetic field produced in this case. [2]
15. State and define S.I unit of magnetic field? [2]
16. How does the strength of magnetic field due to a current carrying conductor depend upon [2]

- i. distance from the conductor?
- ii. current flowing through the conductor?

17. The diagram shows a current-carrying coil passing through a cardboard sheet. Draw three magnetic lines of force on the board. State two factors on which magnitude of magnetic field at the centre depends. [2]

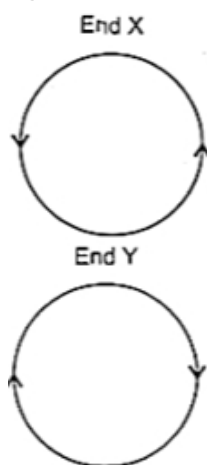


18. When is the force experienced by a current-carrying conductor placed in a magnetic field largest? [2]
19. Complete the following sentences: [2]

- a. A fuse is connected in _____ with the _____ wire.
- b. Higher the current rating _____ is the fuse wire.
- c. A current carrying solenoid behaves like a _____
- d. A current or a moving charge produces a _____ around it

20. Meena draws magnetic field lines of field close to the axis of a current carrying circular loop. As she moves away from the centre of the circular loop she observes that the lines keep on diverging. How will you explain her observation? [2]

21. The directions of current flowing in the coil of an electromagnet at its two ends X and Y are as shown in given figure. [2]



- a. What is the polarity of end X?
- b. What is the polarity of end Y?
- c. Name the rule which you have used to determine the polarities.

22. What are magnetic field lines? List two characteristic properties of these lines. [2]

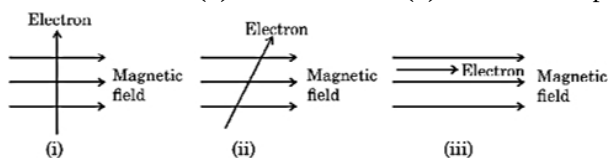
23. Give reasons for the following: [2]

- i. There is either a convergence or a divergence of magnetic field lines near the ends of a current-carrying straight solenoid.
- ii. The current-carrying solenoid when suspended freely rests along a particular direction.
- iii. The burnt-out fuse should be replaced by another fuse of identical rating.

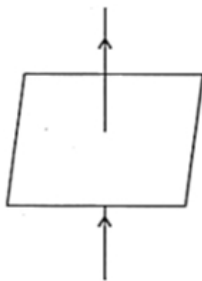
24. State Fleming's left-hand rule. [2]

25. Explain with the help of a labelled diagram the distribution of magnetic field due to a current through a circular loop. Why is it that if a current carrying coil has n turns the field produced at any point is n times as large as that produced by a single turn? [2]

26. What are magnetic field lines? How is the direction of a magnetic field at a point determined? Mention two important properties of magnetic field lines. [2]
27. List three methods of producing magnetic field. [2]
28. List four sources of magnetic field. [2]
29. What is magnetic effect of electric current? [2]
30. i. State the rule used to find the force acting on a current carrying conductor placed in a magnetic field. [2]
 ii. Given below are three diagrams showing entry of an electron in a magnetic field. Identify the case in which the force will be (1) maximum and (2) minimum respectively. Give reason for your answer.

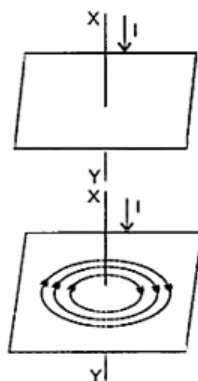


31. a. A straight wire conductor passes vertically through a piece of cardboard sprinkled with iron filings. Copy the diagram and show the setting of iron filings when a current is passed through the wire in the upward direction and the cardboard is tapped gently. Draw arrows to represent the direction of the magnetic field lines. [2]



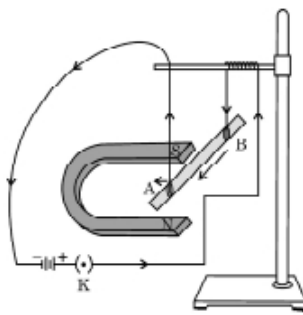
- b. Name the law which helped you to find the direction of the magnetic field lines.
32. The magnetic field in a given region is constant. Draw a diagram to represent it. [2]
33. Kanchi draws magnetic field lines of field close to the axis of a current carrying circular loop. As she moves away from the centre of the circular loop she observes that the lines keep on diverging. How will you explain her observation? [2]
34. What is a solenoid? Draw a diagram to show the pattern of magnetic field around a current carrying solenoid. Name the region of uniform magnetic field. [2]
35. Consider a circular loop of wire lying in the plane of the table. Let the current pass through the loop in the clockwise direction. Apply right-hand rule to find out the direction of the magnetic field inside and outside the loop. [2]
36. Explain why, it is more dangerous to touch the live wire of a mains supply rather than the neutral wire. [2]
37. A house has a main fuse of 5 A rating, 3 bulbs each of 100 W and 8 tube lights each of 40 W are used simultaneously. Calculate : [2]
 i. the current drawn from the mains of 220 V and
 ii. the number of additional bulbs each of the 60 W which can also be lighted?
38. What is meant by short-circuiting ? [2]
39. Draw magnetic field lines around a bar magnet. [2]
40. (a) Define the term current rating of an electric fuse? [2]
 (b) Name the material used to make electric fuse?
 (c) Name two safety measure commonly used in electric circuit and appliances?
41. When does an electric short circuit occur? [2]

42. A fuse is rated 8 A. Can it be used with an appliance of rating 2 kW, 220 V? [2]
43. Compare the permanent magnet and an electromagnet. [2]
44. State the condition in each case of the magnitude of force on a current carrying conductor placed in a magnetic field to be [2]
- zero and
 - maximum
45. When does an electric short circuit occur? [2]
46. Why is the earth pin in a three pin plug thicker and longer than the other two pins? [2]
47. Draw magnetic field lines produced around a straight current carrying conductor passing through a cardboard. How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the conductor? [2]
48. What is the role of fuse used in series with any electrical appliance? Why should a fuse with defined rating not be replaced by one with a larger rating? [2]
49. What kind of magnetic field is produced by a current carrying solenoid? [2]
50. What is an electromagnet? Draw a circuit diagram to show how a soft iron piece can be transformed into an electromagnet. [2]
51. In the diagram XY is a straight conductor carrying current in the direction marked by the arrow. The conductor is held vertical by passing it through a horizontal cardboard sheet. Draw three magnetic lines of force on the board and mark the direction of magnetic field in your diagram. State two factors on which magnitude of magnetic field at a point, depends. [2]



52. State the rule to determine the direction of a [2]
- magnetic field produced around a straight conductor carrying current
 - force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it, and
 - current induced in a coil due to its rotation in a magnetic field.
53. An electric oven of 2 kW power rating is operated at 220 V that has a current rating of 5 A. What do you expect in such a situation? Explain. [2]
54. i. A coil of insulated wire is connected to a galvanometer. What would be observed if a strong bar magnet with its south pole towards one face of the coil is [2]
- moved quickly toward it?
 - moved quickly away from it?
 - held stationary near it?
- ii. Name the phenomena involved.
- iii. State the conclusion based on the observations in (i), (ii) and (iii).

55. A house has a main fuse of 5 A rating. What is the maximum number of 40 W tube lights which can be used at a 220 V supply? [2]
56. Why does a compass needle show deflection when brought near a bar magnet? [2]
57. Name and state the rule to determine the polarity of the two faces of a current carrying circular loop. [2]
58. Write note on magnetic field in human being and animals. [2]
59. A circuit has a fuse of 5 A. What is the maximum number of 100 watt (220 V) bulbs which can be safely used in the circuit? [2]
60.
 - a. What is a solenoid? [2]
 - b. Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows.
61. List the properties of magnetic field lines. [2]
62. List three sources of magnetic fields. [2]
63. A current carrying conductor is placed perpendicular to the uniform magnetic field. What happens to displacement of the conductor if
 - (i) Amount of current increases
 - (ii) If horse shoe magnet is replaced by a weak horse shoe magnet. [2]
64. When is the force experienced by a current-carrying straight conductor placed in a uniform magnetic field? [2]
 - i. Maximum;
 - ii. Minimum?
65. List two methods of producing magnetic fields. [2]
66. Why does a magnetic compass needle pointing North and South in the absence of a nearby magnet get deflected when a bar magnet or a current carrying loop is brought near it? [2]
67.
 - (a) What is a short circuit? [2]
 - (b) What is overloading? How can you avoid overloading?
68. A magnetic compass shows a deflection when placed near a current carrying wire. How will the deflection of the compass get affected if the current in the wire is increased? Support your answer with a reason. [2]
69. Consider a circular loop of wire lying in the plane of the table. Let the current pass through the loop clockwise. Apply the right-hand rule to find out the direction of the magnetic field inside and outside the loop. [2]
70.
 - i. Draw the pattern of magnetic field lines of [2]
 1. a current carrying solenoid
 2. a bar magnet
 - ii. List two distinguishing features between the two fields.
71. As shown in the diagram an aluminium rod **AB** is suspended horizontally between the two poles of a strong horse shoe magnet in such a way that the axis of rod is horizontal and the direction of the magnetic field is vertically upward. The rod is connected in series with a battery and a key. [2]



State giving reason:

- What is observed when a current is passed through the aluminum rod from end B to end A?
- What change is observed in a situation in which the axis of the rod 'AB' is moved and aligned parallel to the magnetic field and current is passed in the rod in the same direction?

OR

Magnetic field is a physical quantity that has both direction and magnitude. How can this statement be proved with the help of magnetic field lines of a bar magnet?

72. Magnetic field in a given region is uniform. Draw a diagram to represent it. [2]
73. Why don't two magnetic lines of force intersect each other ? [2]
74. Which of the following fuse ratings would be suitable for an electric motor of power 3 kW, if it is operated at 220 V supply? (5 A, 10 A, 12 A, 15 A) [2]
75. What are hazards of electricity ? [2]